ComSense Circuits: Mechanistic Interpretability of Commonsense Reasoning Failures in Large Language Models

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Abstract

Large language models routinely fail at commonsense reasoning tasks that seem trivial to humans, yet the internal mechanisms underlying these failures remain poorly understood. We present a systematic mechanistic interpretability study of commonsense reasoning failures in Qwen3-8B, evaluated on the Com2Sense dataset of 2,390 True/False statements organized in complementary pairs. Starting from 611 *asymmetric pairs*—where the model answers one member of a complementary pair correctly and the other incorrectly—we conduct a five-phase investigation combining residual-stream probing, activation patching, head-level analysis, mean-ablation, logit-lens trajectory analysis, and MLP/attention sublayer decomposition. Our central finding is that commonsense reasoning failures are *fundamentally distributed*: no single layer or attention head is sufficient (0% flip rate under activation patching across 18 layers and 160 head-layer combinations) or necessary (maximum 1% flip rate under mean-ablation of 17 candidate heads) for correct commonsense predictions. Probing accuracy plateaus at $\sim$59% even with optimal regularization, and layer-by-layer logit-lens trajectories reveal that 76.5% of incorrect predictions are already directionally wrong at layer 0, with the wrong answer consolidating stably around layer 20. These results contrast sharply with prior mechanistic work on factual recall, where localized circuits with 30–60%+ flip rates are the norm, and suggest that commonsense reasoning requires a fundamentally different diagnostic and repair strategy than surgical circuit-level intervention.

1 Introduction

Large language models achieve impressive scores on many commonsense benchmarks, yet still fail on surprising cases that seem trivial to humans (Davis & Marcus, 2023). Understanding *why* these failures occur—and where in the model they originate—is a central challenge for mechanistic interpretability. Prior work has successfully localized factual recall circuits in GPT-2 and related models (Wang et al., 2023; Geva et al., 2023; Meng et al., 2022), demonstrating that specific attention heads and MLP layers mediate factual associations with patching flip rates of 30–60%+. Commonsense reasoning, however, presents a structurally different challenge: unlike factual associations that can be attributed to specific (subject, relation, object) triples, commonsense judgments require integrating world knowledge, physical intuition, temporal reasoning, and social understanding that may be encoded in a distributed, non-local manner.

We ask three questions:

1. Are commonsense reasoning failures **localized** to specific layers or attention heads, or are they **distributed** across the network?

2. Do failures reflect a **knowledge-retrieval** deficit (the model never encodes the relevant knowledge) or an **inference** failure (knowledge is present but not used correctly)?

3. Are the failures primarily **architectural** (the model cannot represent commonsense), **representational** (it can but doesn't), or **training-induced** (insufficient training signal)?

To answer these questions, we evaluate Qwen3-8B (Qwen Team, 2025) on Com2Sense (Singh et al., 2021), a benchmark of 2,390 True/False statements organized in complementary pairs (e.g., "Rocks make a good pillow" paired with "Rocks are too hard to sleep on"). The model achieves 70.25% overall accuracy, identifying 611 *asymmetric pairs* where it answers one member correctly and the other incorrectly. We use 200 of these pairs—selected for high wrong-answer confidence—as the substrate for mechanistic analysis across five complementary techniques: residual-stream probing, activation patching (layer-level, head-level, and sublayer-level), head-level probing, mean-ablation, and logit-lens trajectory analysis, all implemented via TransformerLens (Nanda & Bloom, 2022).

Our main finding is that commonsense failures are *fundamentally distributed*: patching any single layer or head produces at most $\sim$1.5% prediction flips, probing any individual component achieves only $\sim$59% accuracy, and logit-lens trajectories reveal a complex multi-stage failure process beginning at the earliest layers. This distinguishes commonsense from factual recall and has implications for how alignment-relevant commonsense deficits can be diagnosed and repaired.

2 Background and related work

Mechanistic interpretability. Elhage et al. (2021) introduced the residual stream framework for analyzing transformer computations as a sum of contributions from attention heads and MLP layers. Wang et al. (2023) demonstrated that indirect object identification in GPT-2 Small is mediated by a specific circuit of attention heads. Geva et al. (2023) showed that factual recall in GPT models is localized to specific attention heads and MLP layers, with activation patching achieving 30–60%+ flip rates. Meng et al. (2022) further showed that factual associations can be edited by modifying specific MLP weight matrices. Our work tests whether commonsense reasoning exhibits comparable locality.

Probing classifiers. Probing trains linear classifiers on intermediate representations to measure whether a concept is linearly encoded (Belinkov, 2022). We use probing to measure whether commonsense truth and model correctness are decodable from the residual stream, finding only weak signal ($\sim$59%) that plateaus in late layers.

Logit lens. nostalgebraist (2020) proposed projecting intermediate residual states through the final unembedding to recover layer-by-layer token predictions. We apply this to measure how commonsense judgments evolve across layers, finding a non-monotonic trajectory with early directional bias and late consolidation.

Commonsense benchmarks. Com2Sense (Singh et al., 2021) provides 2,390 True/False statements organized in complementary pairs spanning physical, social, and temporal domains. Unlike HellaSwag (Zellers et al., 2019) or PIQA (Bisk et al., 2020), Com2Sense is designed for *contrastive* evaluation—asymmetric pair performance directly probes reasoning consistency.

3 Methodology

Figure 1 provides an overview of our five-phase experimental pipeline.

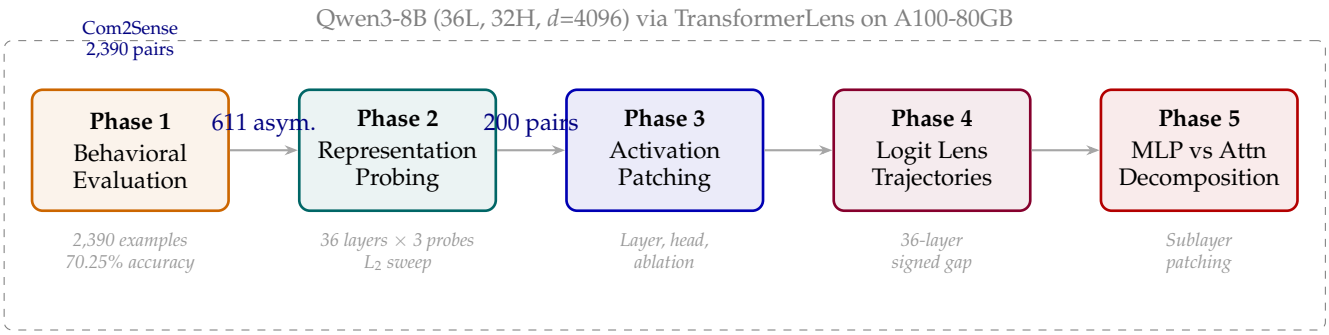

Figure 1: Overview of the five-phase experimental pipeline. Each phase builds on the outputs of the previous: behavioral evaluation identifies asymmetric failure pairs, probing localizes signal to layers 20+, patching and ablation test sufficiency and necessity of individual components, logit lens reveals layer-by-layer prediction trajectories, and sublayer decomposition tests attention vs. MLP contributions.

3.1 Model and dataset

We use Qwen3-8B (36 layers, 32 attention heads per layer, d_{model}=4096, d_{head}=128) loaded via TransformerLens in bfloat16 on an A100-80GB GPU provisioned through Modal (Modal Labs, 2023). We disable Qwen3's chain-of-thought mode (`/no_think`) to obtain clean residual streams without reasoning tokens. For evaluation, we format each statement as:

```
[statement]. Is this True or False? Answer:
```

and compare the logits of the `True` and `False` tokens at the final position (no generation). Com2Sense is sourced from `tasksource/com2sense` on HuggingFace, with complementary pair assignments obtained from the official GitHub mapping at `PlusLabNLP/Com2Sense`.

3.2 Asymmetric pair selection

From 1,195 valid complementary pairs, we identify 611 *asymmetric* pairs where the model answers exactly one member correctly. For mechanistic analysis, we select 200 pairs: 103 from the time domain, 50 temporal, and 47 social, prioritizing pairs with high-confidence wrong answers (mean confidence 0.896, mean logit gap 2.534) to maximize the signal-to-noise ratio of our interventions.

3.3 Representation probing

For each of the 200 pairs (400 examples), we run `run_with_cache()` and extract the residual stream at the final token position (`hook_resid_post`) across all 36 layers, yielding a $[36, 4096]$ tensor per example. We train logistic regression probes with `GroupKFold(n=5)` keeping pairs together to prevent data leakage from near-duplicate complementary sentences. We probe for two labels: (1) model correctness (y=1 if the model's prediction matches ground truth), and (2) ground truth (y=1 if the statement is True). We also test PCA-50 dimensionality reduction and sweep L_2 regularization strength $C \in \{0.001, 0.01, 0.1, 0.5, 1.0, 5.0, 10.0\}$ to control for overfitting in the high-dimensional (d=4096), low-sample (n=400) regime.

3.4 Activation patching

For each of 200 pairs and 18 layers (layers 18–35, the region where probing detects signal), we:

1. Cache the *correct* example's `hook_resid_post` at the final token position.
2. Re-run the *incorrect* example with a hook substituting the cached activation at layer ℓ.

Table 1: Accuracy by domain and scenario on Com2Sense (2,390 examples).

Subset	Accuracy	Count
Overall	70.25%	2,390
Physical	76.27%	805
Social	68.77%	794
Time	66.60%	536
Temporal	63.53%	255
Causal	70.53%	1,198
Comparison	70.02%	804
Comparative	70.03%	387

3. Record the logit change and whether the prediction flips.

We also decompose by sublayer, patching `hook_attn_out` and `hook_mlp_out` separately, for 200 pairs $\times$ 18 layers $\times$ 2 sublayer types $=$ 7,200 forward passes.

3.5 Head-level analysis

Head patching. We patch individual attention head z vectors (`attn.hook_z`) from correct to incorrect examples across the top 5 layers by patching effect (L18, L19, L20, L21, L26)—160 head-layer combinations per pair.

Head probing. We train logistic regression probes (C=0.1) on per-head z vectors (128-dimensional) at the final token position for all 576 heads in layers 18–35 (18 layers $\times$ 32 heads). We also probe at the statement-end position to test whether earlier token positions carry stronger signal.

Mean-ablation. For the 17 most influential heads identified by patching and probing combined (10 from patching: L20.H8, L21.H19, L26.H26, L26.H25, L19.H31, L18.H16, L21.H15, L21.H16, L18.H6, L20.H9; 7 from probing: L32.H20, L34.H18, L28.H11, L24.H11, L28.H25, L32.H12, L35.H23), we compute the mean z vector across all 400 examples for each head, then replace each target head's z output with this mean on the 200 *correct* examples. This tests *necessity*: if a head is essential for correct predictions, removing its unique contribution (replacing with the population mean) should degrade performance.

3.6 Logit lens

We project the cached final-token residual stream through the model's final layer norm and unembedding at every layer, recovering a signed logit gap:

$$\Delta_\ell = \text{logit}[\texttt{True}]_\ell - \text{logit}[\texttt{False}]_\ell$$

To make trajectories comparable across examples with different ground-truth labels, we sign-correct so that positive values indicate the model supports the ground truth and negative values indicate it supports the wrong answer.

4 Results

4.1 Phase 1: Behavioral evaluation

Qwen3-8B achieves **70.25% accuracy** on Com2Sense (1,679/2,390 correct). Table 1 breaks down performance by domain and scenario. The model performs best on physical statements (76.27%) and worst on temporal ones (63.53%), contrary to our initial hypothesis that physical-causal reasoning would be hardest.

Table 2: False-bias in asymmetric failures: percentage of failures where the model incorrectly rejects a True statement.

Domain / Scenario	Failed TRUE	Total	% TRUE
Overall	450	611	73.6%
Social	162	208	77.9%
Physical	125	169	74.0%
Time	111	157	70.7%
Temporal	52	77	67.5%
Social-causal	87	100	87.0%
Physical-causal	73	95	76.8%
Time-causal	57	80	71.2%
Temporal-comparative	22	36	61.1%

Table 3: Phase 2 probing results (original C=1.0 and L_2-corrected).

Probe	Layer	Original	Best C	Corrected
Correctness (no PCA)	32→34	54.75%	0.001	57.50%
Correctness + PCA-50	30→23	61.00%	0.01	58.75%
Ground truth + PCA-50	31→32	60.00%	0.01	59.25%

False-bias. Of the 611 asymmetric pairs, **73.6% involve the model failing on a True statement**—the model systematically defaults to "False" when uncertain. Table 2 shows this bias is pervasive across all domains, peaking at 87.0% in social-causal scenarios. This directional asymmetry is not a generic difficulty effect: it reveals a systematic inference default where the model preferentially rejects statements rather than accepting them under uncertainty.

4.2 Phase 2: Representation probing

Layer-wise probing. Figure 2 (left) shows per-layer probing accuracy across three configurations. All probes are at chance (∼50%) through layers 0–19 and rise sharply from layer 20 onward, peaking in the 23–34 range. Table 3 summarizes the original probe results.

L_2 regularization correction. With 4096-dimensional features and only ∼400 examples, overfitting is a real concern. Figure 2 (right) shows the L_2 sweep results. The raw (no PCA) probe *improves* with stronger regularization (54.75% → 57.50%), confirming that C=1.0 was fitting noise in 4096 dimensions. The PCA-50 probes drop ∼2% (61% → 58.75%), revealing that PCA already provided most of the regularization benefit, but the original results were still slightly inflated. The genuine probing signal is ∼59%—real but weak.

Key finding. The near-identical accuracy of the correctness probe (58.75%) and ground-truth probe (59.25%) after regularization is notable: the model's representation of *what it will predict* is roughly as strong as its representation of *what is actually true*. This means the model does not "know" the correct answer and fail to use it; it genuinely lacks a strong internal representation of commonsense truth.

4.3 Phase 3: Activation patching

Layer-level patching. Figure 3 shows the results of replacing the incorrect example's full residual stream with the correct example's at each layer. Across all 18 layers (18–35), the **prediction flip rate is 0%**—no single layer's activation is sufficient to repair the prediction. All logit changes are negative, ranging from −0.006 (L18) to −0.355 (L35), meaning patching consistently *degrades* predictions. This monotonic degradation reflects distributional mismatch: the correct and incorrect examples are different sentences, and

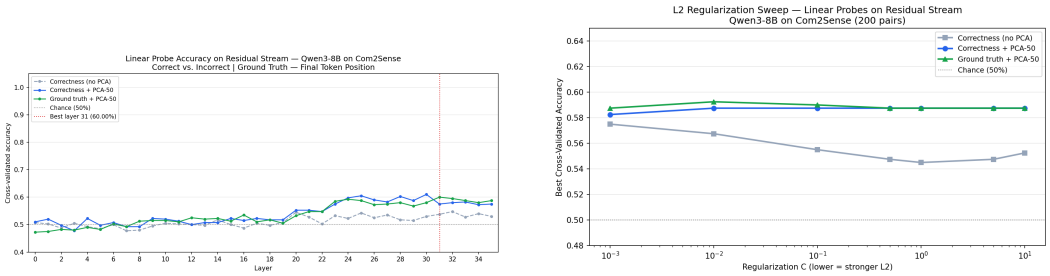

Figure 2: **Left:** Per-layer probing accuracy for three probe configurations. Signal is at chance through layers 0–19 and rises from layer 20. **Right:** L_2 regularization sweep. The corrected probing ceiling is ∼59%.

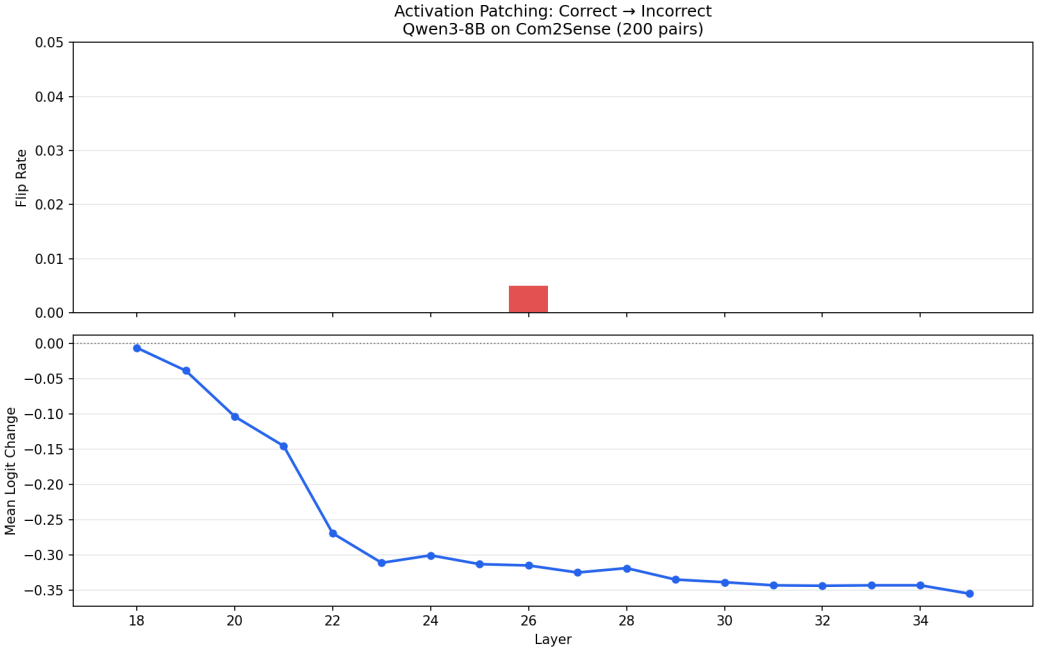

Figure 3: Layer-level activation patching results (layers 18–35). Flip rate is 0% at all layers; logit change is uniformly negative and grows in magnitude with depth, reflecting distributional mismatch from full-stream replacement.

replacing the entire post-layer state creates activations that are inconsistent with what downstream layers expect.

Head-level patching. Figure 4 (left) shows the top 10 attention heads by mean absolute patching effect. The most influential head (L20.H8) moves the logit by 0.147 on average—negligible compared to the mean correct–incorrect logit gap of 2.534. No head produces any prediction flips.

Head-level probing. Figure 4 (right) shows the layer × head probing accuracy heatmap. The best individual head (L32.H20) achieves 59.50%—nearly identical to the full residual-stream probe (∼59%). No head stands out; the signal is weakly but redundantly present across many heads. Crucially, **the most informative heads differ from the most causally influential heads**: the probing leaders (L32, L34) are distinct from the patching leaders (L20, L21), demonstrating that information presence and causal influence are dissociated.

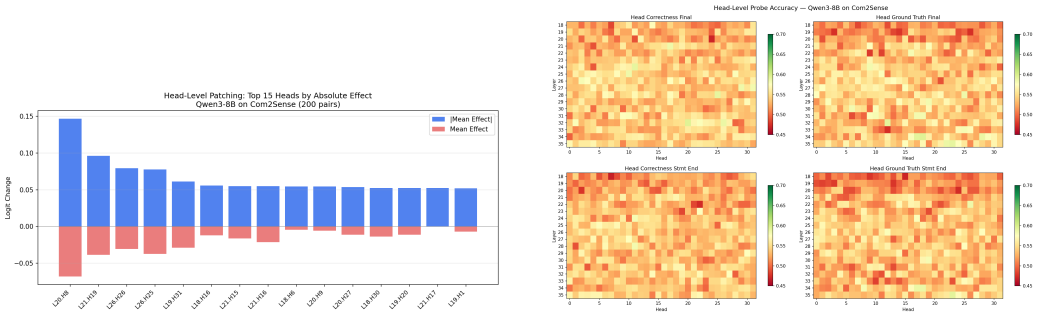

Figure 4: **Left:** Top attention heads by mean absolute patching effect. Maximum effect is 0.147 (L20.H8); no head produces prediction flips. **Right:** Layer $\times$ head probing accuracy heatmap. Best individual head (59.5%) matches full-stream accuracy, indicating the signal is redundantly distributed.

Table 4: Mean-ablation results for the 17 candidate heads (sorted by flip count). Source indicates whether the head was identified by patching or probing.

Head	Flips	Flip%	Mean Δlogit	Std	Source
L26.H25	2	1.0%	$+0.007$	0.153	Patching
L20.H8	1	0.5%	$+0.003$	0.186	Patching
L21.H19	1	0.5%	-0.014	0.126	Patching
L26.H26	1	0.5%	$+0.014$	0.156	Patching
L19.H31	1	0.5%	-0.006	0.096	Patching
L21.H16	1	0.5%	-0.001	0.084	Patching
L28.H25	1	0.5%	-0.004	0.073	Probing
L18.H6	0	0.0%	$+0.028$	0.081	Patching
L32.H20	0	0.0%	$+0.003$	0.067	Probing
L34.H18	0	0.0%	$+0.002$	0.069	Probing
L35.H23	0	0.0%	-0.006	0.073	Probing

Position probing. We also probed at the statement-end token position (last token of the statement, before the prompt suffix). The final token is stronger by $\sim$3%: 60% (layer 31) vs. 57.25% (layer 28). This indicates that the model's commonsense judgment continues to develop through the prompt suffix tokens ("Is this True or False? Answer:"), and is not fully formed by the end of the statement itself.

Mean-ablation. Table 4 shows the results of mean-ablating each of the 17 candidate heads on the 200 correct examples. The maximum flip rate is **1.0%** (L26.H25, 2 flips out of 200), and the maximum absolute logit change is 0.028 (L18.H6)—negligible. No head is necessary for correct commonsense predictions. This result complements the patching findings: patching tests sufficiency (no component is sufficient), while ablation tests necessity (no component is necessary). Together, they establish that commonsense reasoning is genuinely distributed.

4.4 Phase 4: Logit-lens trajectory analysis

The logit-lens analysis fundamentally refines our understanding of failure timing. Figure 5 (left) shows the aggregate signed logit gap trajectories for correct and incorrect examples across all 36 layers. The trajectories reveal a striking three-stage pattern:

1. **Early divergence (layers 0–4):** Correct and incorrect trajectories start separated from layer 0 ($+0.97$ vs. -0.97), already pointing in opposite directions.
2. **Mid-layer convergence (layers 5–19):** Both trajectories partially reverse toward zero, with correct examples briefly turning negative at layers 10–15 and incorrect examples briefly turning positive.

Table 5: Logit-lens failure timing statistics (200 incorrect examples).

Metric	**Value**
Wrong at layer 0	76.5% (153/200)
Mean first-wrong layer	1.61
Mean stable-wrong layer	19.9
Mean stable-correct layer (correct examples)	20.2
Mean final gap (correct examples)	+2.81
Mean final gap (incorrect examples)	−2.53
Final gap by domain (incorrect examples):	
Social	−3.52
Temporal	−2.30
Time	−2.20

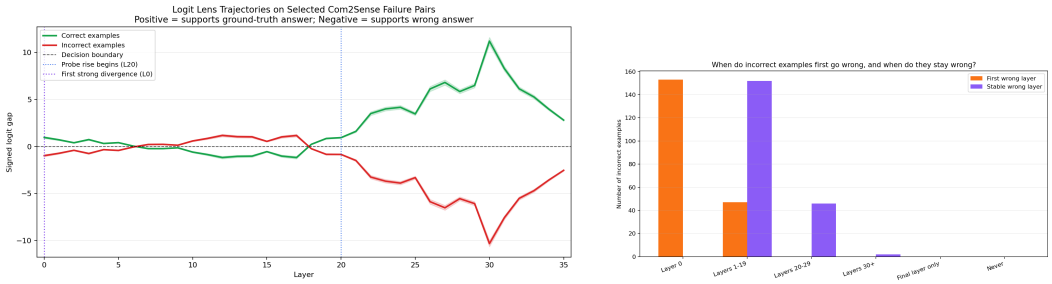

Figure 5: **Left:** Aggregate logit-lens trajectories. Signed logit gap (positive = supports ground truth) for correct (green) and incorrect (red) examples across 36 layers. Trajectories diverge at layer 0, partially converge through mid-layers, then re-diverge strongly from layer 20. **Right:** Distribution of first-wrong (orange) and stable-wrong (blue) layers for the 200 incorrect examples.

3. **Late amplification (layers 20–35):** Trajectories re-diverge sharply from layer 20, peaking at ± 11 around layer 30 before settling to ± 2.5 at the final layer.

Failure timing. Among the 200 incorrect examples, **76.5% (153/200) are already directionally wrong at layer 0**—the signed gap is negative from the first measured layer. Table 5 shows that the mean first-wrong layer is 1.61, but the mean *stable*-wrong layer (where the wrong-answer bias becomes sustained) is 19.9—matching the layer range where probing first detects signal. This two-stage pattern—early bias followed by late consolidation—is more nuanced than either a pure retrieval failure or a pure inference failure.

Domain-level trajectories. Figure 6 shows logit-lens trajectories broken down by domain for incorrect examples. Social failures end in the strongest wrong-state (−3.52), consistent with the highest false-bias rate (77.9%). Temporal failures show the weakest final wrong-state (−2.30) but the lowest behavioral accuracy (63.53%), suggesting that temporal failures are more evenly distributed between confident wrong answers and uncertain wrong answers.

4.5 Phase 5: MLP vs. attention sublayer decomposition

To test whether commonsense signal concentrates in attention or MLP sublayers—as has been reported for PIQA-style tasks in GPT-2 Small—we decompose layer patching into separate `hook_attn_out` and `hook_mlp_out` interventions.

Table 6 shows selected results. Unlike full `resid_post` patching (uniformly negative), sublayer patching shows *mixed signs*—some layers produce positive logit changes, others negative. This sign variability reveals that the monotonically negative `resid_post` results were an artifact of compounding distributional mismatch, not evidence of layer-specific causal importance.

Table 6: Selected MLP vs. attention sublayer patching results (layers 18–35).

Layer	Attn flip%	Attn Δlogit	MLP flip%	MLP Δlogit
18	0.0%	−0.011	0.0%	−0.003
20	0.0%	−0.081	0.0%	−0.014
22	0.0%	−0.156	0.0%	−0.149
25	**1.5%**	**+0.163**	0.0%	+0.007
28	0.0%	+0.003	0.0%	−0.045
32	0.0%	−0.011	0.0%	−0.012
35	0.0%	−0.018	**1.5%**	**+0.133**

Neither sublayer dominates. The only prediction flips occur at L25 attention (1.5% flip rate, +0.163 mean logit change—the largest positive intervention in the entire study) and L35 MLP (1.5% flip rate, +0.133). Attention effects are larger in magnitude in middle layers (L19–L27), while MLP effects match or exceed attention only at L22–L23 and in late layers (L30–L35). This differs from the MLP-dominance pattern reported for GPT-2 Small on PIQA, likely reflecting both architectural differences (Qwen3-8B uses grouped-query attention) and task differences (Com2Sense requires multi-hop causal/temporal reasoning vs. PIQA's single-step physical intuition).

5 Discussion

5.1 Answers to research questions

Q1: Localized or distributed? Distributed. The convergent evidence across all experiments is unambiguous:

- Layer-level patching: 0% flip rate across 18 layers (no layer is sufficient).
- Head-level patching: 0% flip rate across 160 combinations (no head is sufficient).
- Sublayer patching: maximum 1.5% flip rate (neither attention nor MLP is sufficient).
- Mean-ablation: maximum 1.0% flip rate across 17 candidate heads (no head is necessary).
- Head-level probing: best head (59.5%) matches full-stream accuracy (∼59%), indicating the signal is redundantly spread rather than concentrated.

This contrasts sharply with factual recall findings. Geva et al. (2023) report flip rates of 30–60%+ for factual associations. Wang et al. (2023) identify a specific circuit for indirect object identification. Commonsense reasoning—with its demand for multi-hop world knowledge integration—does not admit such localization.

Q2: Knowledge retrieval or inference failure? Early bias with late consolidation. The probing results initially suggested a purely late-onset inference failure (signal emerging at layer 20, chance before that). Logit-lens analysis reveals a more nuanced two-stage process: 76.5% of incorrect examples are already directionally wrong at layer 0, suggesting an *early representational bias*—possibly reflecting training data priors such as the observed false-bias (the model defaults to rejection under uncertainty). However, these early biases are not stably committed until ∼layer 20, and the non-monotonic trajectory (with mid-layer reversals) shows the network iteratively revises its judgment before final consolidation.

This is inconsistent with either a pure retrieval failure (the model never had the knowledge) or a pure inference failure (wrong computation on correct representations). Instead, the model builds an early weak bias toward the wrong answer, then consolidates it through late-layer computation.

Q3: Architectural, representational, or training-induced? Representational, with possible training-data contributions. The evidence most strongly supports a representational deficit:

Table 7: Comparison with prior mechanistic interpretability studies.

Study	Task	Method	Flip rate
Wang et al. (2023)	IOI (GPT-2)	Circuit patching	High
Geva et al. (2023)	Factual recall	Layer patching	30–60%+
Meng et al. (2022)	Factual editing	ROME (MLP)	High
This work	Commonsense	Layer patching	**0%**
This work	Commonsense	Head patching	**0%**
This work	Commonsense	Sublayer patching	**1.5%**
This work	Commonsense	Mean-ablation	**1.0%**

- Probing accuracy of $\sim$59% means the residual stream carries only weak signal about commonsense truth. The model never builds a strong enough internal representation to reliably distinguish correct from incorrect reasoning.
- The architecture *can* encode commonsense distinctions (59% $>$ 50%), just weakly. A purely architectural limitation would more likely show 50% (no signal at all) or concentrate signal in specific components.
- The sublayer decomposition shows that attention can transiently encode useful signal (L25 attention: $+0.163$), but not consistently enough for reliable judgment. The architecture is not the bottleneck; the *strength* of the representation is.

We cannot fully distinguish representational from training-induced with our methods. If the model never encountered sufficient commonsense reasoning examples during training, the representational deficit would be a consequence of training data limitations. Distinguishing these would require comparing models trained on different data mixtures, which is beyond our scope.

5.2 Why patching fails for commonsense

The near-zero patching flip rates require explanation, since patching reliably works for factual recall. We identify two complementary reasons:

Distributional mismatch. The correct and incorrect examples in each pair are *different sentences* with different token sequences (e.g., "Rocks make a good pillow" vs. "Rocks are too hard to sleep on"). Replacing one sentence's residual stream with another's creates activations inconsistent with the tokens downstream layers processed. This is fundamentally different from factual recall patching, where the same sentence is used with different corruptions.

Genuine distribution of signal. Even the cleaner sublayer and head-level interventions (which create less distributional mismatch) produce negligible effects. The signal is genuinely too distributed and too weak to be captured by any single-component intervention.

5.3 Comparison with prior work

Table 7 places our findings in the context of prior mechanistic studies. The near-zero flip rates we observe are dramatically lower than those reported for factual tasks, establishing that commonsense reasoning is mechanistically distinct from factual recall.

5.4 Implications

Surgical repair is ineffective. Given that no single component is sufficient or necessary, targeted patching cannot repair commonsense failures in the way it can for factual recall. Approaches like ROME (Meng et al., 2022), which edit specific MLP weight matrices to correct factual errors, are unlikely to generalize to commonsense. Commonsense failures require addressing the underlying *representational* weakness rather than a specific computational bottleneck.

Two-stage failure model. Our findings suggest a revised model of commonsense failure: an *early-bias, late-consolidation* mechanism. Initial directional biases—possibly reflecting training data priors such as the observed false-bias (the model defaults to "False" for uncertain causal/temporal judgments)—are amplified and committed by late-layer computation. The non-monotonic logit-lens trajectories show the network iteratively revises its judgment rather than computing it at a single location, further supporting the distributed interpretation.

Implications for alignment. If commonsense failures are distributed and representationally diffuse, they cannot be fixed by localizing and editing specific circuits. This suggests that improving commonsense reliability likely requires changes to the *training distribution*—specifically, more high-quality commonsense examples that build stronger representational encodings—rather than post-hoc surgical intervention.

5.5 Limitations

This study is limited to a single model (Qwen3-8B) and a single dataset (Com2Sense). The 200-pair subset, while sufficient for reliable statistical patterns, may not capture all failure modes. Probing accuracy of $\sim$59% is real but weak; stronger probes (e.g., nonlinear probes, token-averaged representations) might reveal more structure. We do not ablate full attention layers or MLP blocks, only individual heads; block-level ablation might reveal stronger effects. Causal attribution from activation patching is limited by distributional mismatch between different input sentences; path patching (Conmy et al., 2023), which controls for this mismatch, could reveal additional structure. Finally, we cannot fully distinguish representational from training-induced failures without comparing models trained on different data mixtures.

6 Conclusion

We have presented a systematic mechanistic interpretability study of commonsense reasoning failures in Qwen3-8B, combining five complementary techniques across 200 asymmetric complementary pairs from Com2Sense. Our central finding is that commonsense failures are *fundamentally distributed*: no single layer, attention head, or sublayer is sufficient (0–1.5% flip rates) or necessary (maximum 1% flip rate under ablation) for correct commonsense predictions.

Logit-lens analysis reveals that failures involve an *early-bias, late-consolidation* process: 76.5% of incorrect predictions are already directionally wrong at layer 0, but the wrong answer typically stabilizes around layer 20. Probing accuracy plateaus at $\sim$59%, indicating weak but genuine signal that is redundantly spread across the residual stream rather than concentrated in identifiable components.

These findings challenge the assumption that commonsense failures can be diagnosed and repaired through the same localized circuit-finding methodology that has proven successful for factual recall. Commonsense reasoning appears to require a fundamentally different treatment: strengthening the underlying representational encoding through training, rather than patching specific computational components after the fact.

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A Additional figures

B Full sublayer patching results

C Full ablation results

D MLP vs. attention sublayer visualization

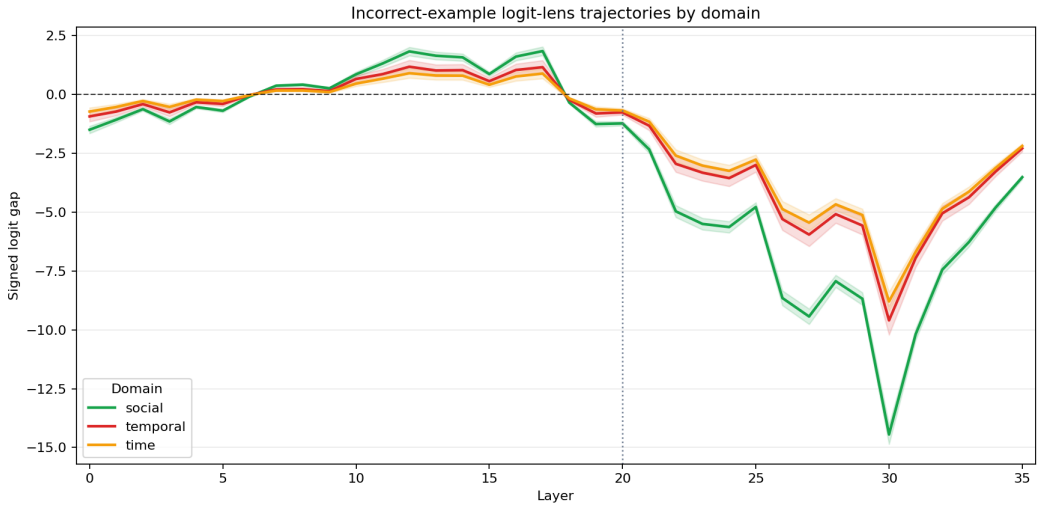

Figure 6: Logit-lens trajectories broken down by domain for incorrect examples. Social failures show the most extreme final wrong-state (-3.52), while temporal failures show the weakest (-2.30).

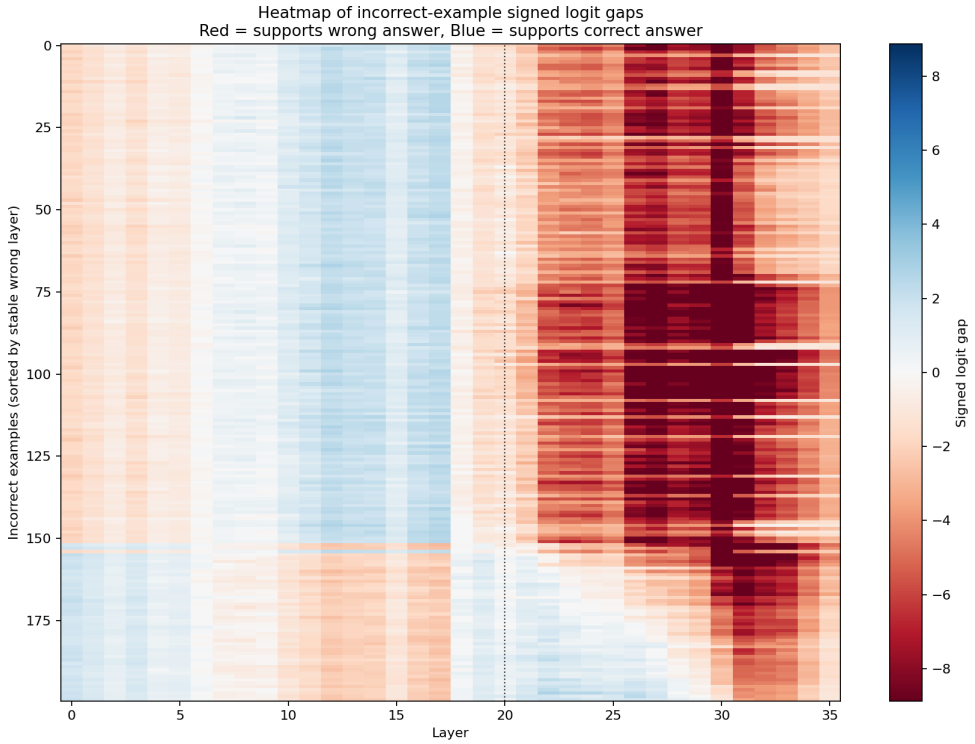

Figure 7: Heatmap of signed logit gaps for all 200 incorrect examples across 36 layers. Rows are individual examples, columns are layers. Most examples are red (wrong direction) from early layers; a minority start blue (correct direction) and transition to red around layer 20.

Table 8: Complete MLP vs. attention sublayer patching results for all 18 layers (18–35).

Layer	Attn flip%	Attn Δlogit	MLP flip%	MLP Δlogit
18	0.0%	−0.011	0.0%	−0.003
19	0.0%	−0.037	0.0%	−0.011
20	0.0%	−0.081	0.0%	−0.014
21	0.0%	−0.063	0.0%	+0.009
22	0.0%	−0.156	0.0%	−0.149
23	0.0%	−0.061	0.0%	−0.146
24	0.0%	−0.041	0.0%	+0.077
25	1.5%	+0.163	0.0%	+0.007
26	0.0%	−0.090	0.0%	−0.072
27	0.0%	−0.060	0.0%	−0.051
28	0.0%	+0.003	0.0%	−0.045
29	0.0%	−0.055	0.0%	−0.051
30	0.0%	−0.066	0.0%	−0.083
31	0.0%	−0.019	0.0%	+0.023
32	0.0%	−0.011	0.0%	−0.012
33	0.0%	−0.032	0.0%	−0.002
34	0.0%	−0.030	0.0%	−0.016
35	0.0%	−0.018	1.5%	+0.133

Table 9: Complete mean-ablation results for all 17 candidate heads, sorted by absolute mean logit change.

Head	Flips	Flip%	Mean Δ	Median Δ	Std	Source
L18.H6	0	0.0%	+0.028	0.0	0.081	Patching
L18.H16	0	0.0%	+0.023	0.0	0.080	Patching
L26.H26	1	0.5%	+0.014	0.0	0.156	Patching
L21.H19	1	0.5%	−0.014	0.0	0.126	Patching
L26.H25	2	1.0%	+0.007	0.0	0.153	Patching
L28.H11	0	0.0%	+0.006	0.0	0.075	Probing
L19.H31	1	0.5%	−0.006	0.0	0.096	Patching
L35.H23	0	0.0%	−0.006	0.0	0.073	Probing
L28.H25	1	0.5%	−0.004	0.0	0.073	Probing
L21.H15	0	0.0%	−0.003	0.0	0.098	Patching
L20.H8	1	0.5%	+0.003	0.0	0.186	Patching
L32.H20	0	0.0%	+0.003	0.0	0.067	Probing
L32.H12	0	0.0%	+0.003	0.0	0.075	Probing
L34.H18	0	0.0%	+0.002	0.0	0.069	Probing
L24.H11	0	0.0%	+0.001	0.0	0.078	Probing
L20.H9	0	0.0%	+0.001	0.0	0.088	Patching
L21.H16	1	0.5%	−0.001	0.0	0.084	Patching

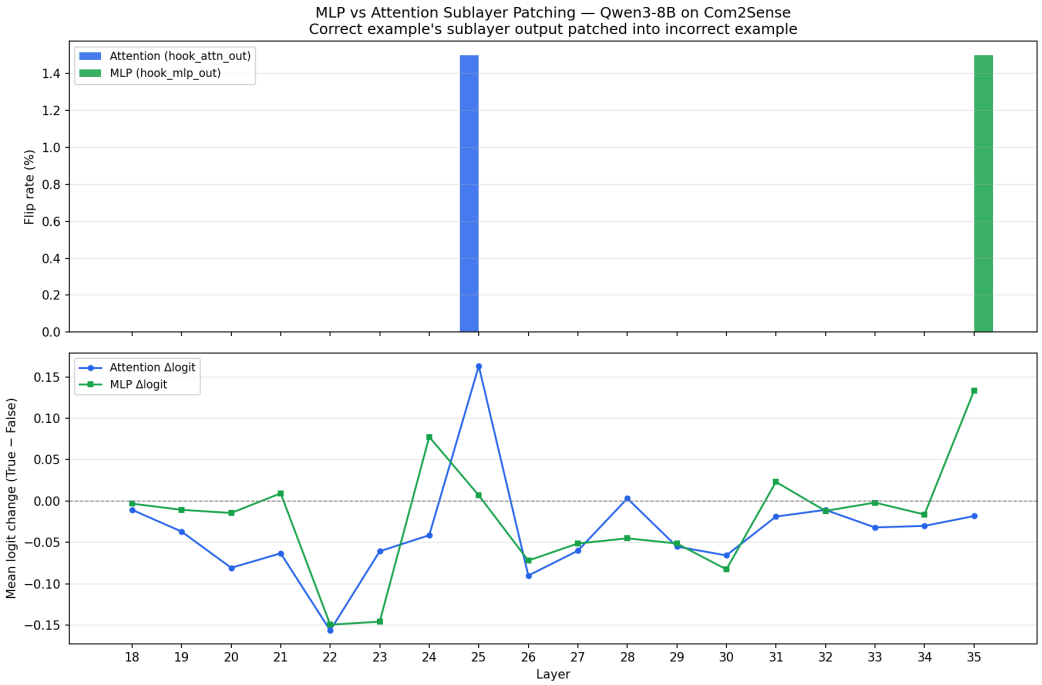

Figure 8: MLP vs. attention sublayer patching results across all 18 layers. Unlike full-stream patching (Figure 3), sublayer effects have mixed signs, revealing that the monotonically negative `resid_post` results were an artifact of distributional mismatch rather than evidence of causal importance.

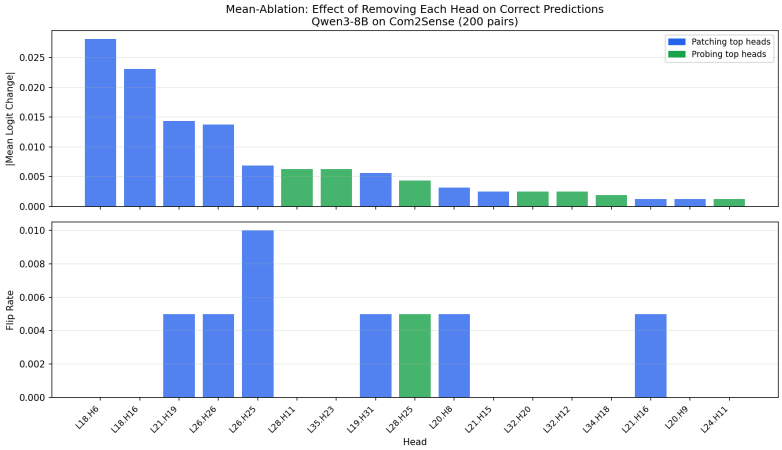

Figure 9: Mean-ablation flip rates for the 17 candidate attention heads. Maximum flip rate is 1.0% (L26.H25, 2/200 correct predictions flipped). No head is necessary for correct commonsense predictions.